

Data visualization and modelling

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2- Permutation tests

Overview

- 1- An example: two sample test.
- 2- Paired test.
- 3- Correlation test.
- 4- Advanced examples.

An example: two sample test.

To compare teaching methods, 20 school children were randomly assigned to one of two groups. The following are the test results:

conventional	65	79	90	75	61	85	98	80	97	75
new	90	98	73	79	84	81	98	90	83	88

Are the two teaching methods equivalent in result?

Resampling methods: a practical guide to data Analysis. Phillip I. Good (2006). Chapter 3.

- **Distributional pattern unknown.**
- **Small sample sizes.**
- **What do we want to compare? Means?, variances?, all?**

Procedure for a permutation (randomization) test :

1. Define your null hypothesis and the alternative.
2. Consider a statistic which is large (small) if your null hypothesis is not true, and small (large) if it is true.
3. Create a new data set consisting of your data, randomly rearranged. Exactly how it is rearranged depends on your null hypothesis.
4. Calculate your statistic for this new data set and compare it to your true value.
5. Repeat steps 3 and 4 many times (preferably several hundred).
6. If your true statistic is greater (lower) than 95% of the random values, then you can reject the null hypothesis at $p < 0.05$.

For our example:

1- H_0 : conventional (X_c) and new (X_n) are equivalent (belong to the same population), that is, X_c and X_n are copies of the same random variable.

In H_1 we want to summarize that “new is better than conventional”. However there are several possibilities to do this:

$$H_1: E(X_n) > E(X_c)$$

$$H_1: P(X_n > X_c) > 0.5 \text{ (or } 0.6 \text{ ?)}$$

$$H_1: \text{median}(X_n) > \text{median}(X_c)$$

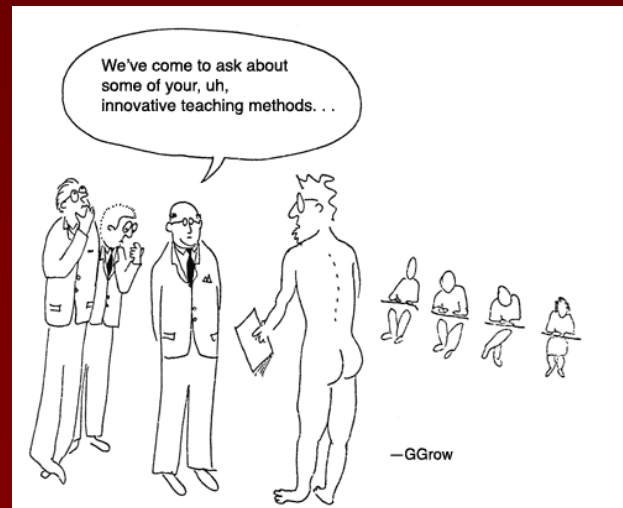
We choose: $H_1: E(X_n) > E(X_c)$

2- An appropriate statistic could be the difference of means: $\text{mean}(\text{new}) - \text{mean}(\text{conventional}) = 5.9$

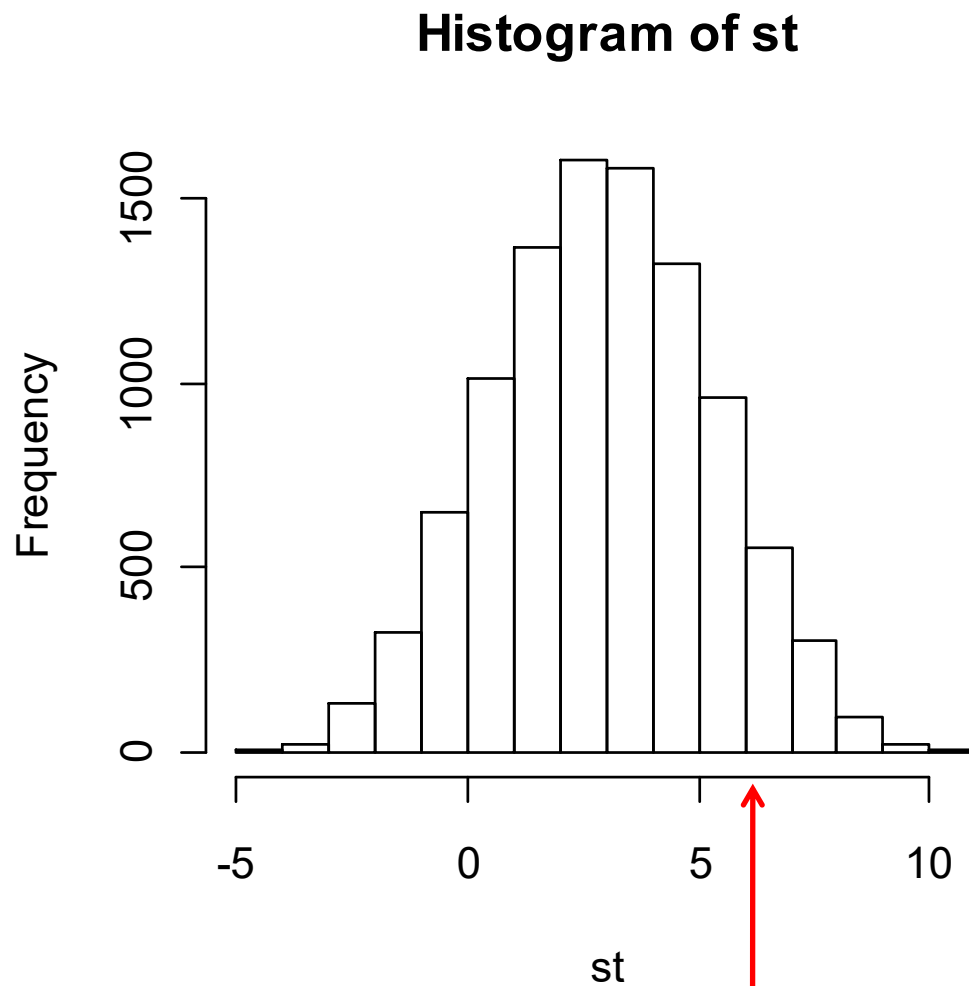
3- New data sets consisting of your data, randomly rearranged: there are $20!$ permutations. We can not be exhaustive: only an approximated test will be possible.

Code in R

```
nr=10000 #number of rearrangements to be examined
st<-numeric(nr)
conv=c(65, 79, 90, 75, 61, 85, 98, 80, 97, 75)
new = c (90, 98, 73, 79, 84, 81, 98, 90, 83, 88)
n1=length (new);n2=length (conv)
total<-n1+n2
sttrue= mean(new)-mean(conv)
cnt= 0 #zero the counter
#Put both sets of observations in a single vector
vect = c(new, conv)
for (i in 1:nr){
d= sample (vect,n1+n2)
ne<-d[1:n1]
a<-n1+1;co<-d[a:total]
st[i]<-mean(ne)-mean(co)
if (st[i] > sttrue)cnt=cnt+1 }
cnt/nr #pvalue
[1] 0.1085
```



```
hist(st)  
sttrue  
[1] 5.9
```



H_0 is not rejected.

5.9

Precision of the p-value

Let p = true p-value (if we do *all possible rearrangements*)

We estimate p by means of a sample proportion.

Consequently we can calculate a $(1-\alpha)\%$ confidence interval of p using the classical expression,

$$\hat{p} \pm z_{1-\alpha/2} \sqrt{\frac{\hat{p} (1 - \hat{p})}{n}}$$

In our example, a 95% confidence interval is
 0.1085 ± 0.006

A paired test

Suppose that we have n pairs of observations,

$(X_1, Y_1), \dots, (X_n, Y_n)$

1- The null hypothesis is that X and Y are interchangeable for each pair. Define the alternative.

2- Consider an appropriate statistic.

3- Randomly flip the pairs. (For each pair, toss a fair coin. If heads, switch X and Y ; if tails, do not switch.)

4- Compare the observed statistic to the distribution of the statistic when the pairs are flipped at random.

An example

We are using a new treatment and we want to know if it decreases the systolic pressure of the patients.

The following results have been taken over 12 individuals:

120 128 124 131 130 131 118 127 140 132 128 125
140 141 135 137 126 118 130 132 126 129 127 135

(the values in red mean “before the treatment”).



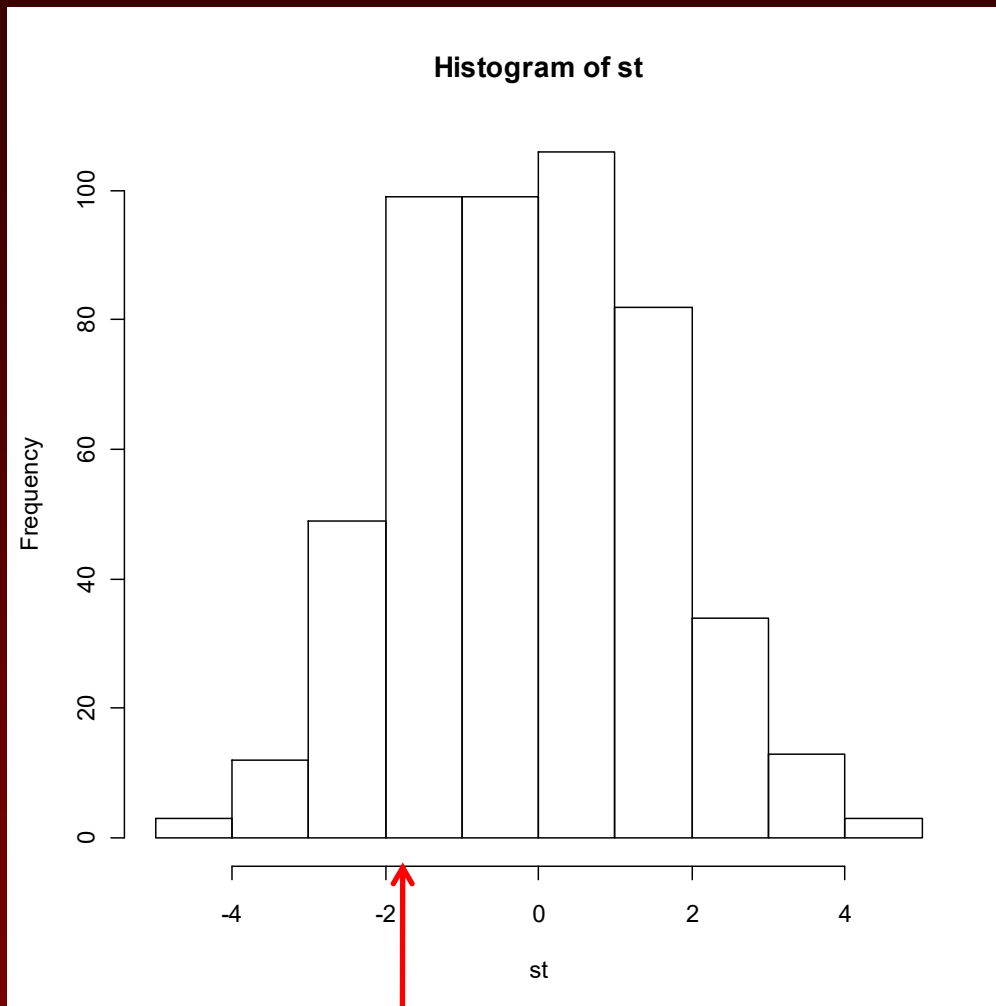
The alternative hypothesis is, $H_1: \mu_b > \mu_a$
“The systolic pressure before the treatment is greater than after”.

- A simple test statistic for the example data is the mean of the paired differences. For our data, the value of this statistic is **-1.833** .
- We have **$2^{12}=4096$** rearrangements . Small n: look at all possible flips. Large n: look at a sample of such flips.
- In order to perform an exact test (all possible flips), the function **intToBin()** implemented in the package R.utils (you have to download it from CRAN) could be useful.

Code in R: approximate test.

```
before= c(120,124,130,118,140,128,140,135,126,130,126,127)
after=c(128,131,131,127,132,125,141,137,118,132,129,135)
diff=before-after
nr=500 #number of rearrangements to be examined
st=numeric(nr)
sttrue= mean(diff)
n=length(diff)
stat = numeric(n)
cnt= 0 #zero the counter
for (i in 1:nr){
  for (j in 1:n) {stat[j]=ifelse(runif(1) < 0.5, diff[j], -diff[j])}
  st[i]=mean(stat)
  if (st[i] < sttrue)cnt=cnt+1}
cnt/nr # p-value
[1] 0.128
```

```
hist(st)  
sttrue  
[1] -1.8333
```



-1.833

H_0 is not rejected.

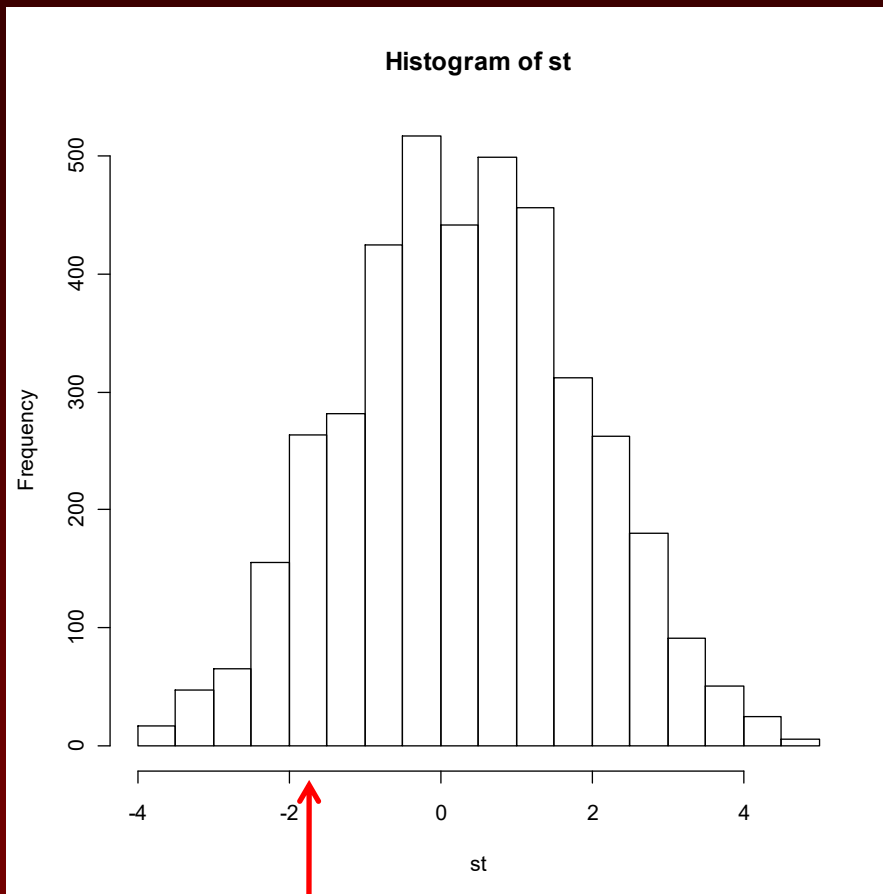
The exact test

```
before= c(120,124,130,118,140,128,140,135,126,130,126,127)
after=c(128,131,131,127,132,125,141,137,118,132,129,135)
diff=before-after
sttrue= mean(diff)
n=length(diff)
nf=2^n-1
st=numeric(nf)
stat = numeric(n)
for (i in 0:nf){
  rearr=intToBin(i)
  nn=nchar(rearr)
  u=strrep("0", n-nn)
  rearr=paste(u,rearr,sep="")
  for (j in 1:n) {stat[j]=ifelse(substr(rearr,j,j)==0, diff[j], -diff[j])}
  st[i]=mean(stat)}
length(st[st<=sttrue])/(2^n) # p-value
[1] 0.1523438
```

```
hist(st)
```

```
sttrue
```

```
[1] -1.8333
```



-1.833

H_0 is not rejected.

A correlation test

Suppose that we have n pairs of observations,

$(X_1, Y_1), \dots, (X_n, Y_n)$

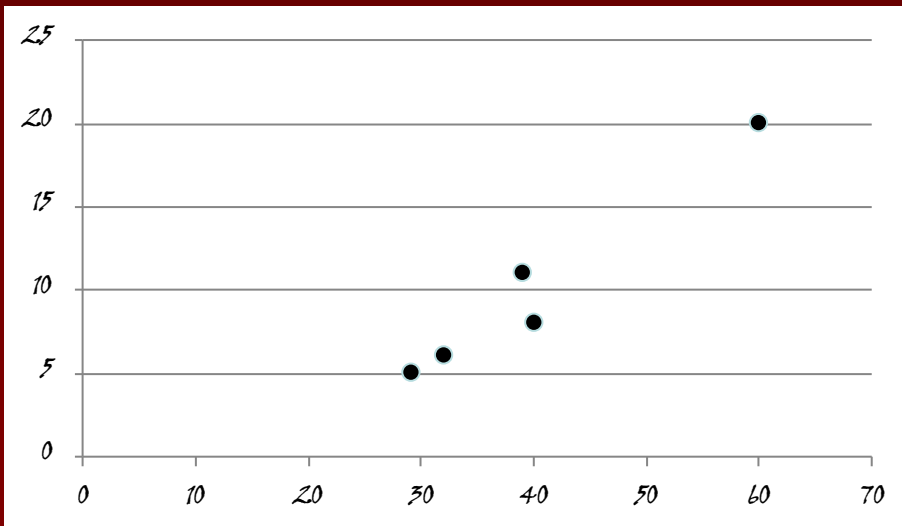
- 1- The null hypothesis is that X and Y are not correlated, that is, we could interchange the X_i between them, constructing new equivalent pairs. The more usual alternative is to assume a linear correlation.
- 2- Consider an appropriate statistic, like Pearson correlation coefficient.
- 3- Permute randomly the Y column leaving the X fixed.
- 4- Compare the observed statistic to the distribution of the statistic when the Y column is permuted.

An example

The first row is chest circumference (in inches) of five subjects. Let us call this X. The second row is the respective total volumes of air that can be breathed in and out in one minute (in liters) for the same five subjects. Let us call this Y.

X: 39, 29, 60, 40, 32

Y: 11, 5, 20, 8, 6



The alternative hypothesis

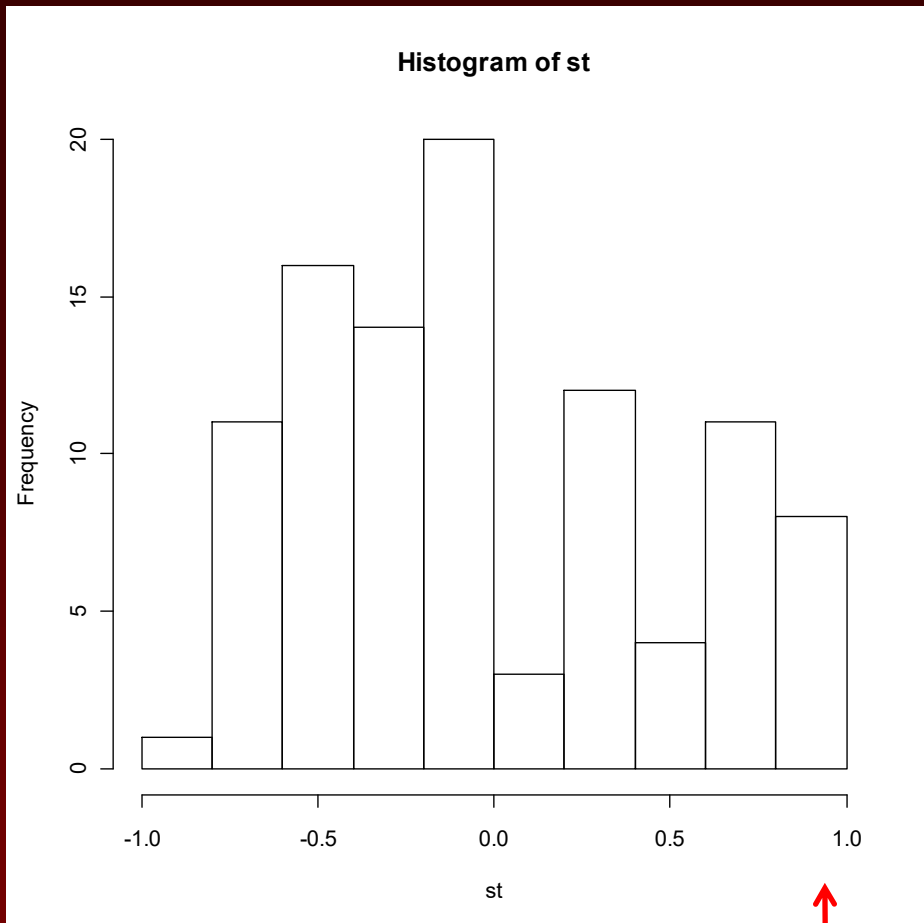
is, $H_1: \rho > 0$

“Chest circumference and volume of air are positively correlated”.

Code in R: approximate test.

```
x= c(39,29,60,40,32)
y=c(11,5,20,8,6)
nr=100 #number of rearrangements to be examined
st=numeric(nr)
sttrue= cor(x,y)
n=length(y)
cnt=0
for (i in 1:nr){
d= sample (y,n)
st[i]<-cor(d,x)
if (st[i] > sttrue)cnt=cnt+1 }
cnt/nr #pvalue
[1] 0.01
```

```
hist(st)
sttrue
[1] 0.978
```



H_0 is rejected.

In order to perform an exact test, the function `permn()` from the package `combinat` could be useful.

0.978

The exact test

```
x= c(39,29,60,40,32)
y=c(11,5,20,8,6)
sttrue= cor(x,y,method="spearman")
n=length(y)
nr=fact(n) #number of rearrangements to be examined
st=numeric(nr)
cnt=0
d=permn(y)
for (i in 1:nr){
  st[i]<-cor(d[[i]],x,method="spearman")
  if (st[i] >=sttrue)cnt=cnt+1 }
cnt/nr #pvalue
```

[1] 0.025

Advanced examples.

Consider the following example concerning the number of ants consumed by two sizes of lizards over each of four months.

	Small	Large
June	13 242 105	182 21 7
July	8 59 20	24 312 68
August	515 488 88	460 1223 990
Septem.	18 44 21	140 40 27

We want to study the effects due to the month and the size of the lizard in the intake.

This is the data
set (lizards.txt):

```
X SIZE MONTH
13 small june
242 small june
105 small june
182 large june
21 large june
7 large june
8 small july
59 small july
20 small july
24 large july
312 large july
68 large july
515 small augu
488 small augu
88 small augu
460 large augu
1223 large augu
990 large augu
18 small sept
44 small sept
21 small sept
140 large sept
40 large sept
27 large sept
```

This is the program in R for performing an
ANOVA two-way:

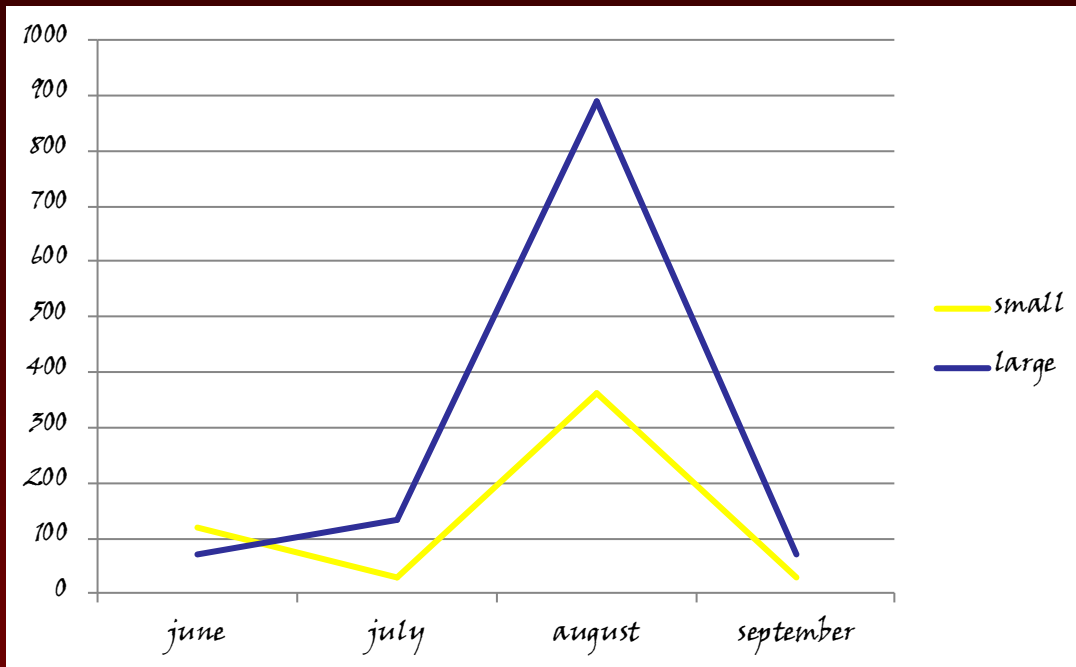
```
data<-read.table("lizards.txt",header=TRUE)
x<-data$X
month<-factor(data$MONTH)
size<-factor(data$SIZE)
fit<-lm(x ~ size + month + size:month)
anova(fit)
```

Analysis of Variance Table

Response: x

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
size	1	146172	146172	4.4699	0.05055 .
month	3	1379495	459832	14.0615	9.487e-05 ***
size:month	3	294009	98003	2.9969	0.06171 .
Residuals	16	523222	32701		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.'
0.1 ' ' 1



A plot of the means indicates a possible interaction due to a change in the behavior of the lizards in June: the average intake of the “small” is grater than the “large”.

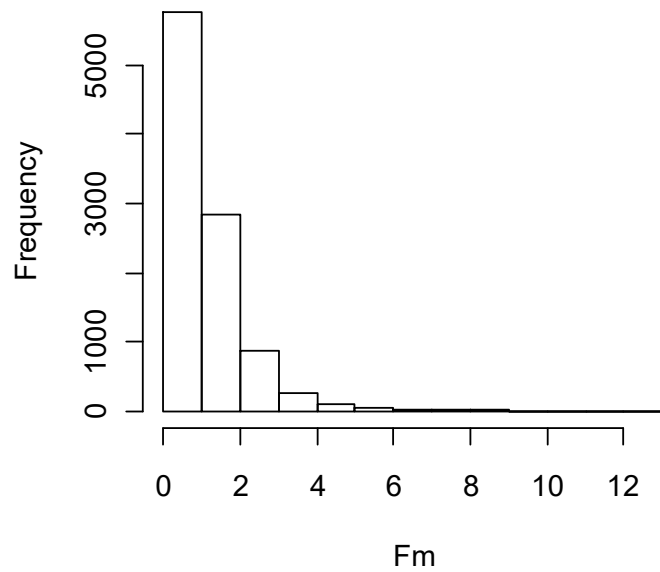
However, here the data are counts and consequently the normality assumption could be removed. We can perform a permutation test using the F-statistics of the ANOVA for testing the principal effects and also the interaction.

One way to permute our data would be to randomize them over all cells in the experiment. For example, we could take our 24 values, shake them up in a hat, and write them down in whatever order they came out of the hat next to the columns that contain information on Size and Months. We would calculate the F for our three effects, store those values away, and then repeat this procedure another 10000 times.

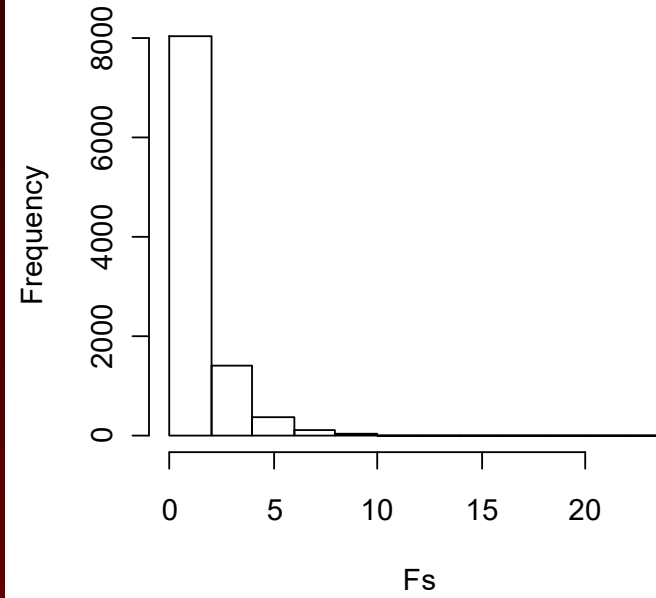
Code in R: approximate test.

```
fit<-lm(x ~ size + month + size:month)
b<-anova(fit)
Fstrue<-b$"F value"[1] ; Fmtrue<-b$"F value"[2]
Fltrue<-b$"F value"[3]
nr=10000 #number of rearrangements to be examined
Fs=numeric(nr);Fm=numeric(nr);Fl=numeric(nr)
for (i in 1:nr){
  newx<- sample(x, 24)
  fit<-lm(newx ~ size + month + size:month)
  b<-anova(fit)
  Fs[i]<-b$"F value"[1] ; Fm[i]<-b$"F value"[2]
  Fl[i]<-b$"F value"[3] }
length(Fs[Fs >= Fstrue])/nr
length(Fm[Fm >= Fmtrue])/nr
length(Fl[Fl >= Fltrue])/nr
```

Histogram of Fm



Histogram of Fs

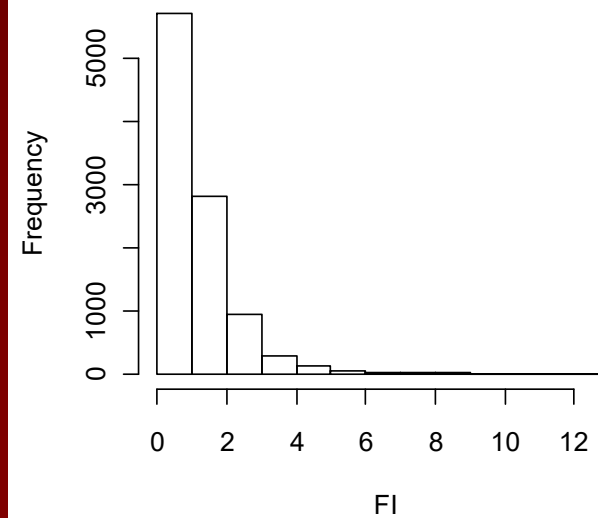


Fs: [1] 0.0416

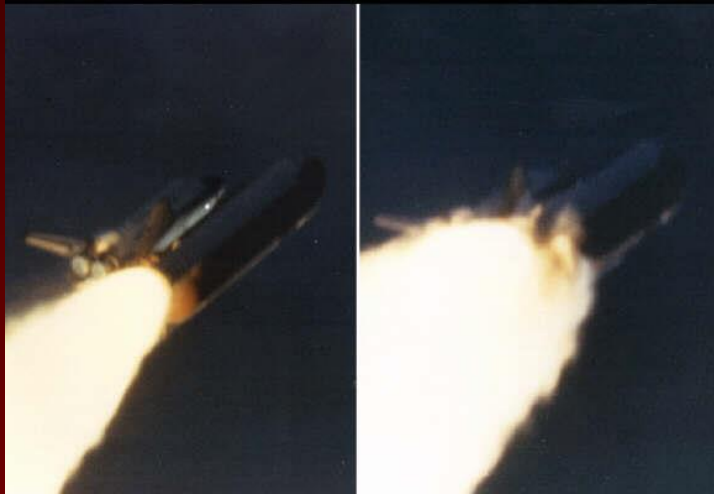
Fm: [1] 0

FI: [1] 0.0519

Histogram of FI



Science needs statistical modelling



In 01/28/1986 the Challenger was launched from Kennedy spatial center in Florida.

73 seconds after the launch, the Challenger exploded and its 7 crew members died.



<https://www.youtube.com/watch?v=9maWclatweM>

LA VANGUARDIA

08001 - BARCELONA
Miércoles, 29 de enero de 1986
Número 37.393

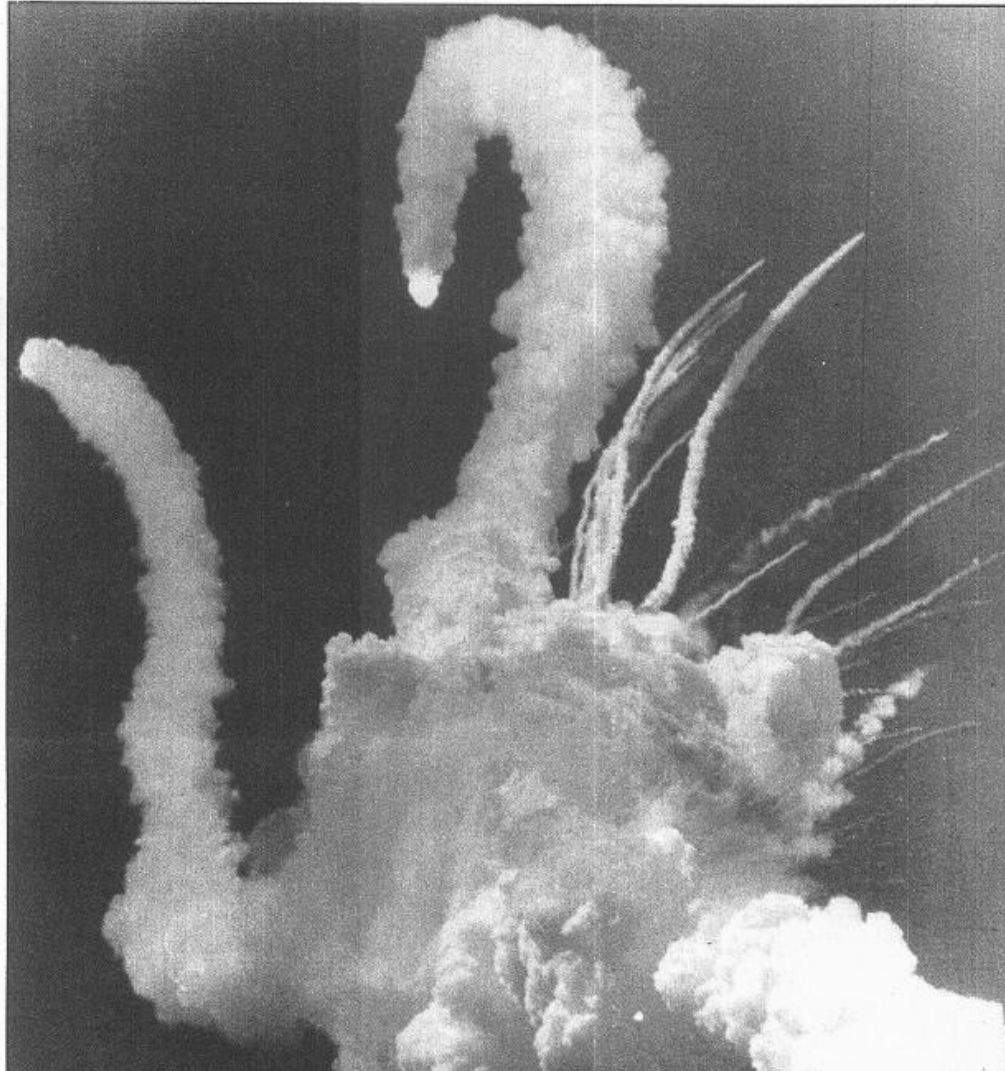
FUNDADA EN 1881
POR DON CARLOS Y DON BARTOLOMÉ GODÓ

Redacción y Administración: Pelayo, 28
Teléfono: 301-54-54. Télex: 54.530 y 54.781
Precio de este ejemplar: **60 pts.**

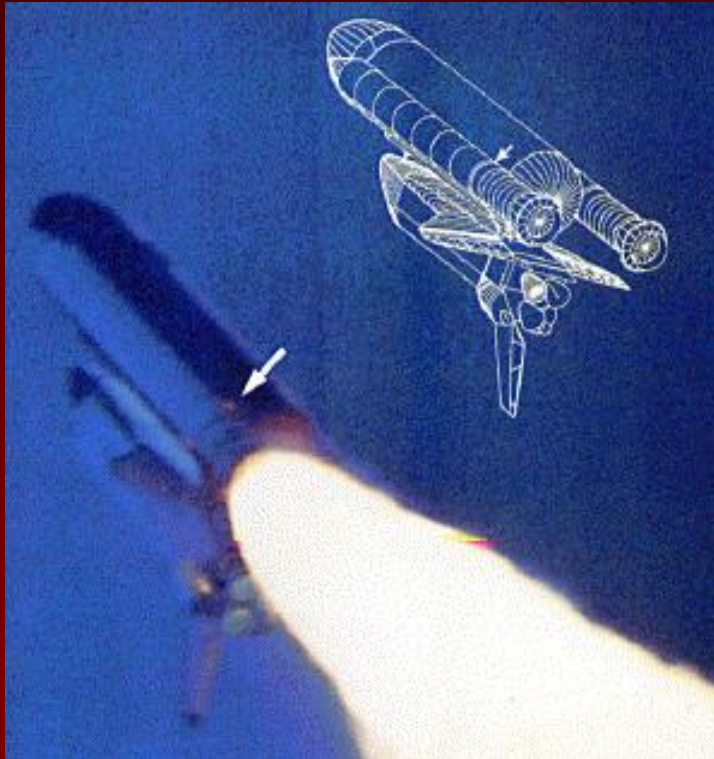
La décima misión del "Challenger" acabó en tragedia



El transbordador espacial norteamericano "Challenger", en el momento de salir de su rampa



What was wrong with Challenger?



An O-ring seal in its right solid rocket booster failed at liftoff.

The failure of the O-ring was attributed to the low temperature on the day of launch (-0.6 C).

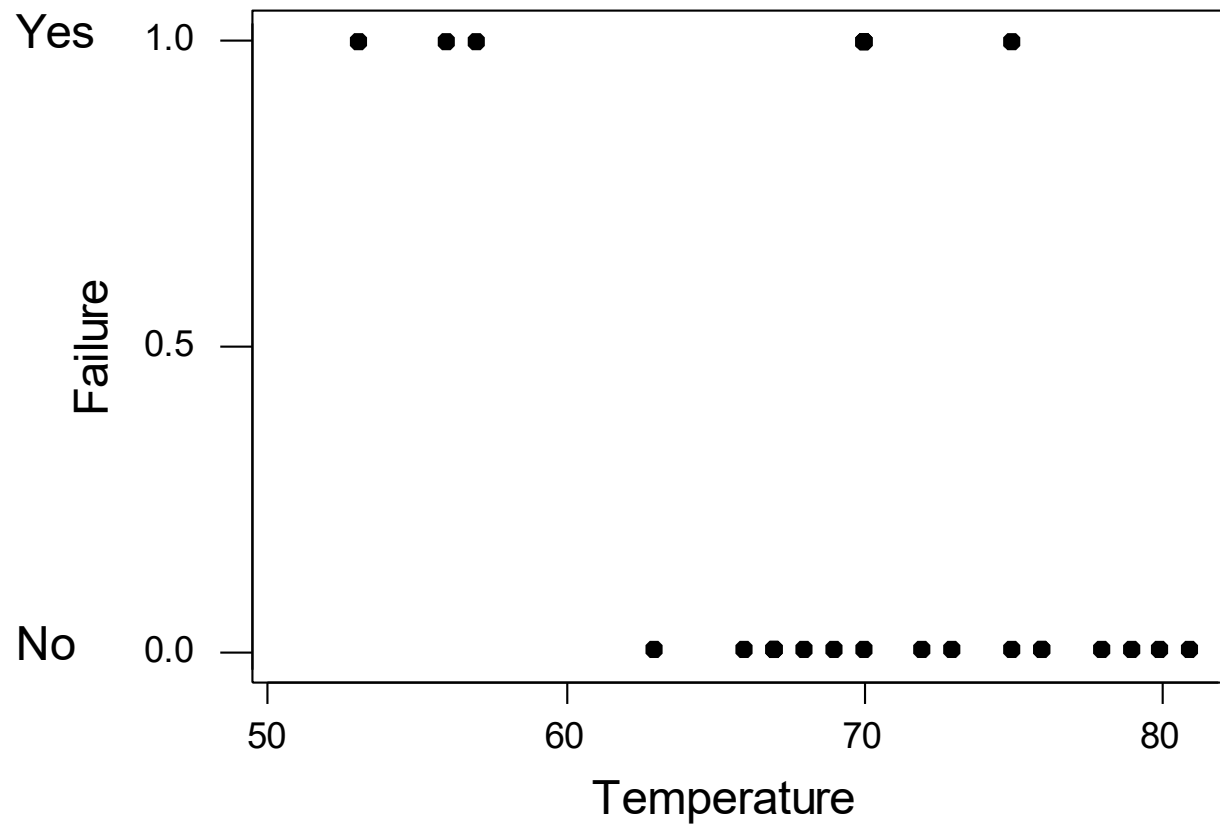
(The rubber of the O-rings is less flexible with low temperatures.)

An appropriate statistical analysis could have avoided the tragedy.

- It was the launch number 25.
- After each launch, the rocket motors were recovered from the ocean.
- 23 previous shuttle flights were launched at temperatures between 53 F and 81 F. In 7 of them, the O-rings presented some damages.

NASA did not detect any relation between O-rings failures and environmental temperature, in spite of the suggestions of Thiokol. They recommended a minimum temperature of 53 F (11.7 C) for a good performance of the O-rings.

Plot of Failure versus Temperature



A simple logistic regression could be done:

```
temp<-c(66,70,69,68,67,72,73,70,57,63,70,78,67,53,67,75,70,81,76,79,75,76,58)
y<-c(0,1,0,0,0,0,0,0,1,1,1,0,0,1,0,0,0,0,0,0,1,0,1)
```

```
logit<-glm(y~temp,family=binomial)
```

```
summary(logit)
```

Call:

```
glm(formula = y ~ temp, family = binomial)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.0611	-0.7613	-0.3783	0.4524	2.2175

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	15.0429	7.3786	2.039	0.0415 *
temp	-0.2322	0.1082	-2.145	0.0320 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 28.267 on 22 degrees of freedom
Residual deviance: 20.315 on 21 degrees of freedom
AIC: 24.315

Number of Fisher Scoring iterations: 5

The day of the accident the temperature was 31 F (-0.6 C).

The probability of failure can be estimated using the logistic regression fit,

$$\hat{p} = \frac{e^{15.043 - 0.232 \cdot 31}}{1 + e^{15.043 - 0.232 \cdot 31}} = 0.9996$$

Consequently, the probability of failure is about 99.9%. The launch must have been delayed !!

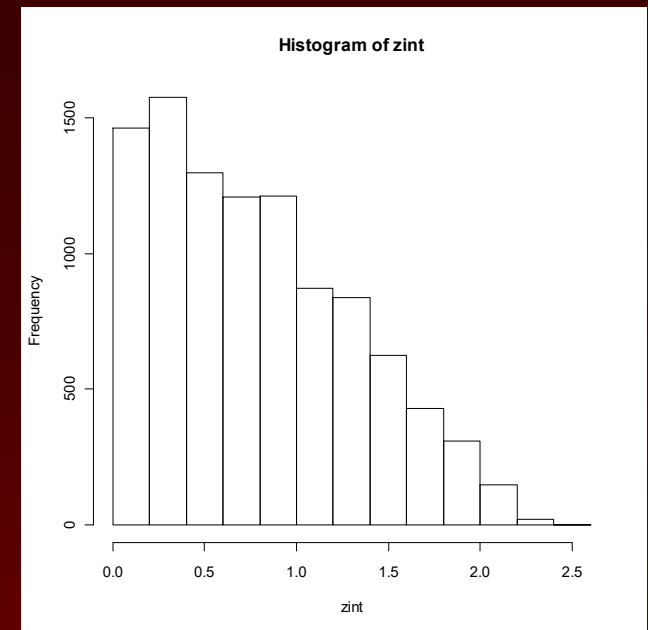
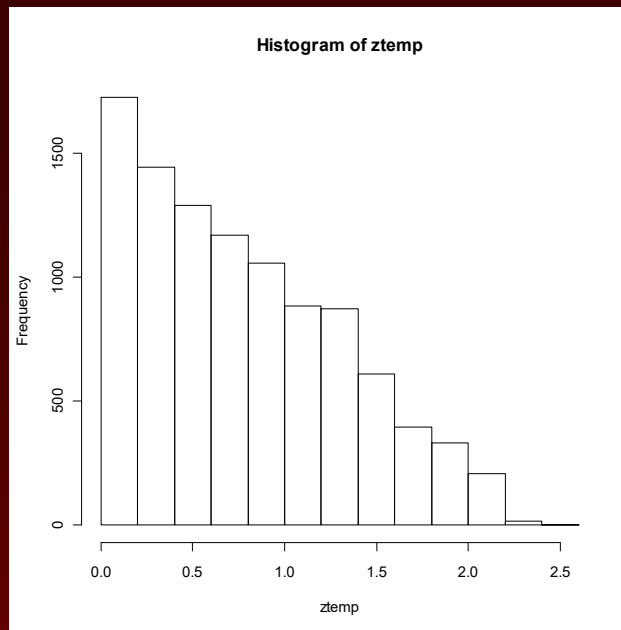
However, the p-values obtained by the logistic regression are asymptotic, because they are based in a maximum likelihood approach.

We can perform a permutation test using the z-value statistics of the logistic regression.

We can rearrange our data and permute randomly the Y column leaving the temperatures fixed. Then, we will calculate the z-value for the two coefficients, store the values, and then repeat this procedure another 10000 times.

Code in R: approximate test.

```
temp<-c(66,70,69,68,67,72,73,70,57,63,70,78,67,53,67,75,70,81,76,
79,75,76,58)
y<-c(0,1,0,0,0,0,0,0,1,1,1,0,0,1,0,0,0,0,0,0,1,0,1)
logit<-glm(y~temp,family=binomial)
a<-summary(logit)
zinttrue<-abs(a$coefficients[1,3])
ztemptrue<-abs(a$coefficients[2,3])
nr=10000 #number of rearrangements to be examined
zint=numeric(nr);ztemp=numeric(nr)
for (i in 1:nr){
  newy<- sample(y, 23)
  logit<-glm(newy~temp,family=binomial)
  a<-summary(logit)
  zint[i]<-abs(a$coefficients[1,3])
  ztemp[i]<-abs(a$coefficients[2,3])}
length(zint[zint >= zinttrue])/nr
length(ztemp[ztemp >= ztemptrue])/nr
```



Now the approximated
p-values are,

Intercept: [1] 0.0103

Temp: [1] 0.0048

Note that the effect of Temperature is more significant.

A randomness test for time series

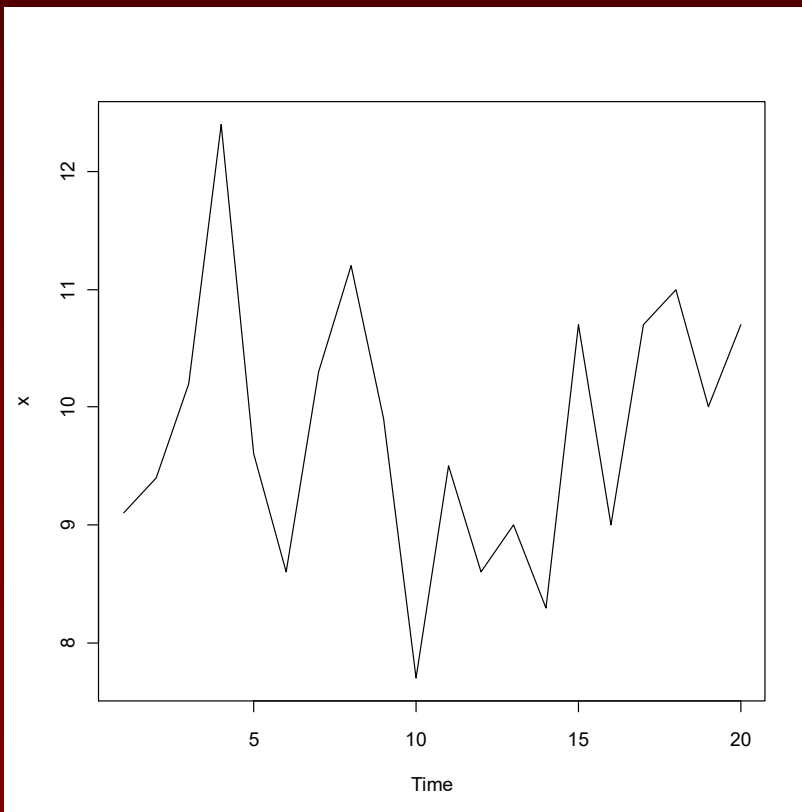
Suppose that we have a time series of n observations, $X_1, \dots, X_n$

- 1- The null hypothesis is that the observations follow a white noise of the form $X_i = \mu + Z_i$, where Z_i are iid such that $E(Z_i)=0$.
- 2- We assume that under the null hypothesis the observations are interchangeable. We can consider simple appropriate statistics like the first autocorrelation coefficient, the Box-Ljung statistic,...
- 3- Permute randomly the values of the time series.
- 4- Compare the observed statistic to the distribution of the statistic when the observations are permuted.

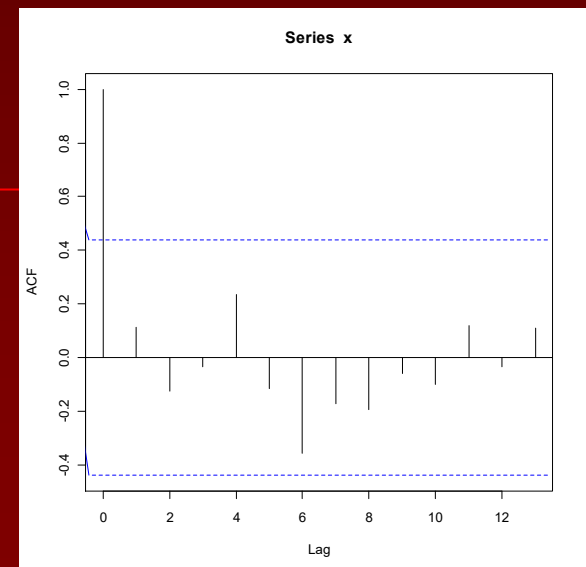
An example

Consider the following 20 observations of the weekly production of a company:

9.1, 9.4, 10.2, 12.4, 9.6, 8.6, 10.3, 11.2, 9.9, 7.7, 9.5, 8.6, 9.0, 8.3, 10.7, 9.0, 10.7, 11.0, 10.0, 10.7



```
x<-c(9.1,9.4,10.2,12.4,9.6,8.6,  
10.3,11.2,9.9,7.7,9.5,8.6,9.0,  
8.3,10.7,9.0,10.7,11.0,10.0,10.7)  
ts.plot(x)  
acf(x)
```

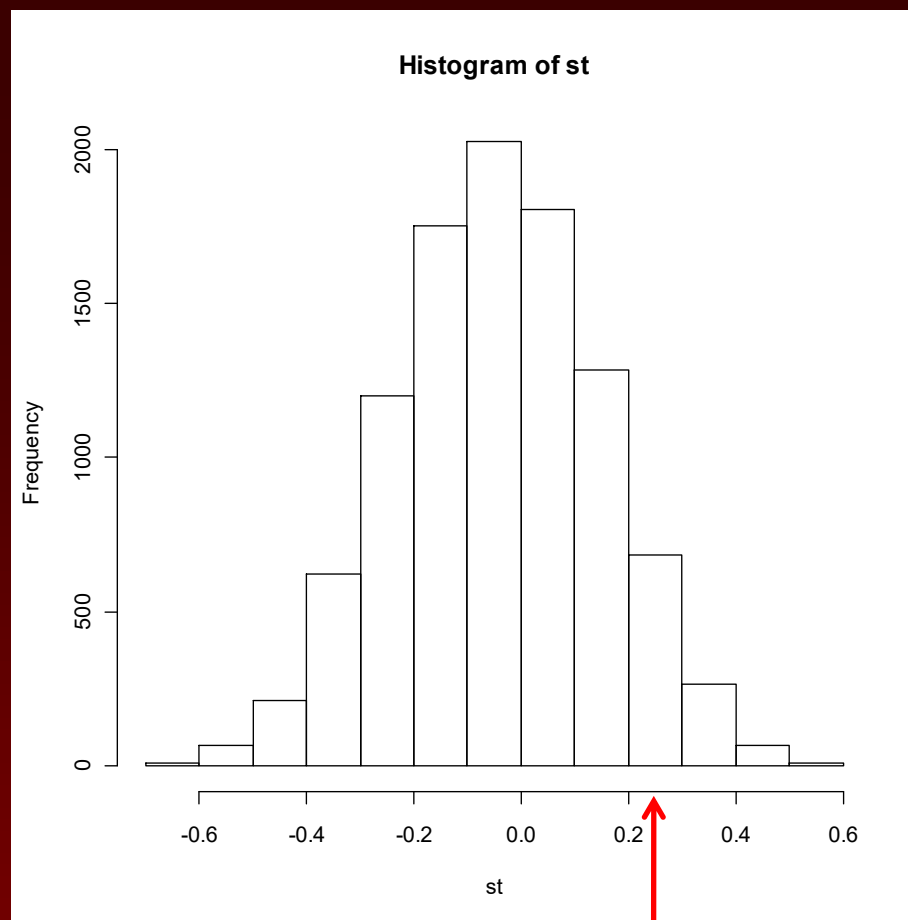


Code in R: approximate test.

```
x<-c(9.1,9.4,10.2,12.4,9.6,8.6,10.3,11.2,9.9,7.7,9.5,8.6,9.0,8.3,  
      10.7,9.0,10.7,11.0,10.0,10.7)  
nr=10000 #number of rearrangements to be examined  
st=numeric(nr)  
sttrue=acf(x)$acf[2]  
n=length(x)  
for (i in 1:nr){  
  d= sample (x,n)  
  st[i]<-acf(d)$acf[2]}  
length(st[st >=sttrue])/nr  
[1] 0.2221
```

But using \$acf[5], that is the autocorrelation coefficient of order 4, we obtain a p-value 0.071.

```
hist(st)
sttrue
[1] 0.235
```



H_0 is not rejected.

0.235

Another example

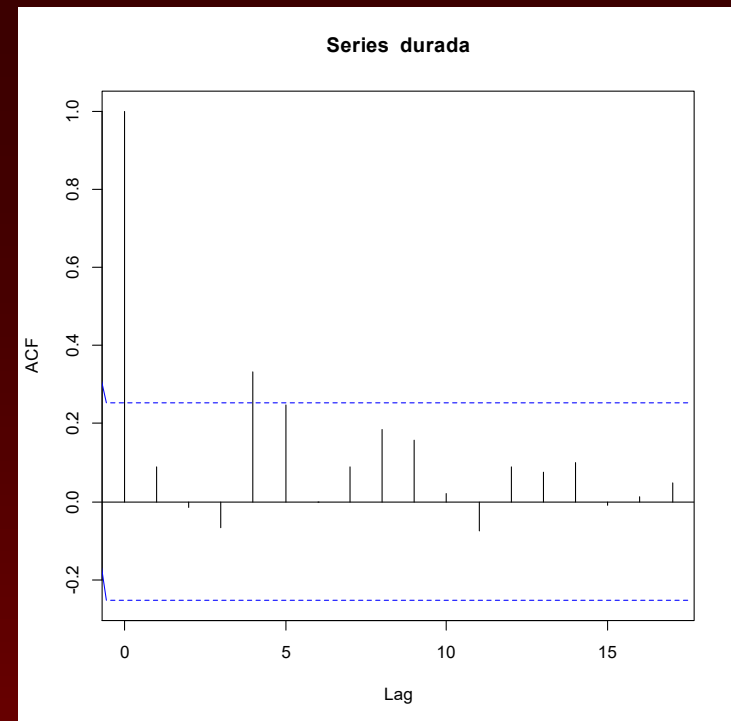
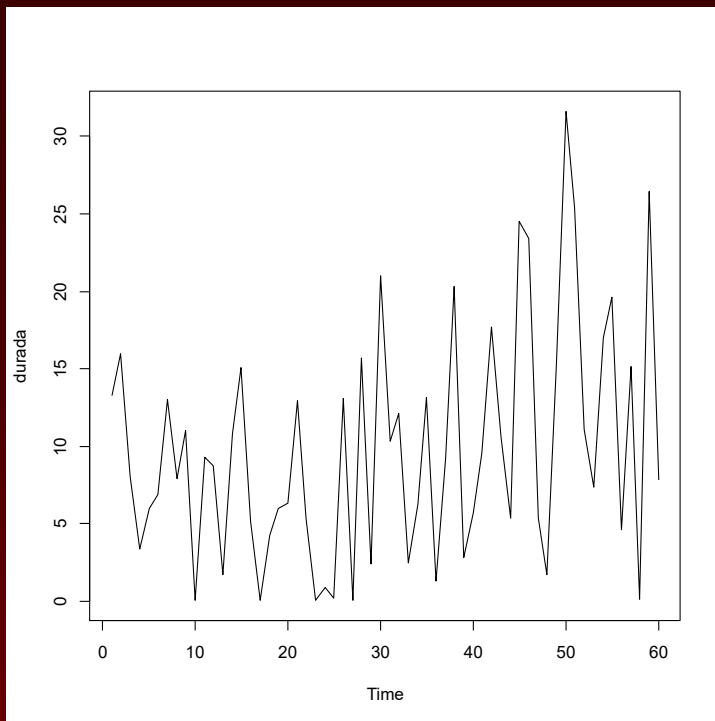
Consider the following observations corresponding to the duration of the reigns of the last 60 popes:

	Start	Stop
1	11/11/1417	20/2/1431
2	03/03/1431	23/2/1447
3	06/03/1447	24/3/1455
4	08/04/1455	06/08/1458
5	19/8/1458	15/8/1464
6	30/8/1464	26/7/1471
7	09/08/1471	12/08/1484
8	29/8/1484	25/7/1492
9	11/08/1492	18/8/1503
10	22/09/1503	18/10/1503
11	31/10/1503	21/02/1513
12	09/3/1513	1/12/1521
13	9/1/1522	14/9/1523
14	26/11/1523	25/9/1534
15	13/10/1534	10/11/1549
16	7/2/1550	29/3/1555
17	9/4/1555	1/5/1555
18	23/5/1555	18/08/1559
19	26/12/1559	9/12/1565
20	7/1/1566	1/5/1572
21	13/5/1572	10/4/1585
22	24/4/1585	27/8/1590
23	15/9/1590	27/9/1590
24	5/12/1590	16/10/1591

.....

[1]	13.27583847	15.97809719	8.04928131	3.32922656	5.99041752	6.90212183
[7]	13.01026694	7.90417522	11.01437372	0.07118412	9.31143053	8.73100616
[13]	1.67830253	10.83093771	15.07734428	5.13620808	0.06023272	4.23819302
[19]	5.95482546	6.31348392	12.90896646	5.34154689	0.03285421	0.86242300
[25]	0.16974675	13.08966461	0.07118412	15.70431211	2.40657084	20.98015058
[31]	10.31074606	12.12320329	2.47227926	6.23134839	13.14442163	1.32238193
[37]	9.21013005	20.31759069	2.83093771	5.73305955	9.57152635	17.70841889
[43]	10.57905544	5.34428474	24.53388090	23.43326489	5.37166324	1.67008898
[49]	15.32648871	31.64681725	25.40725530	11.04449008	7.38672142	17.01026694
[55]	19.60574949	4.59685147	15.12662560	0.09034908	26.46132786	7.86310746



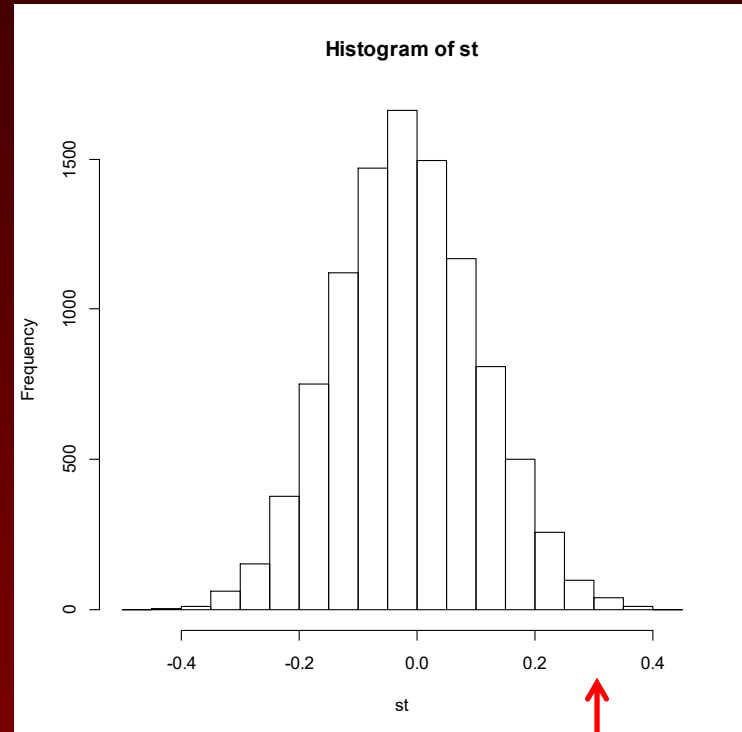


Note a possible significant autocorrelation coefficient of order 4.

Code in R: approximate test.

```
nr=10000 #number of rearrangements to be examined
st=numeric(nr)
sttrue=acf(durada)$acf[5]
n=length(durada)
for (i in 1:nr){
  d= sample (durada,n)
  st[i]<-acf(d,plot=F)$acf[5]}
length(st[st >=sttrue])/nr
[1] 0.0023
```

How, can it be explained???



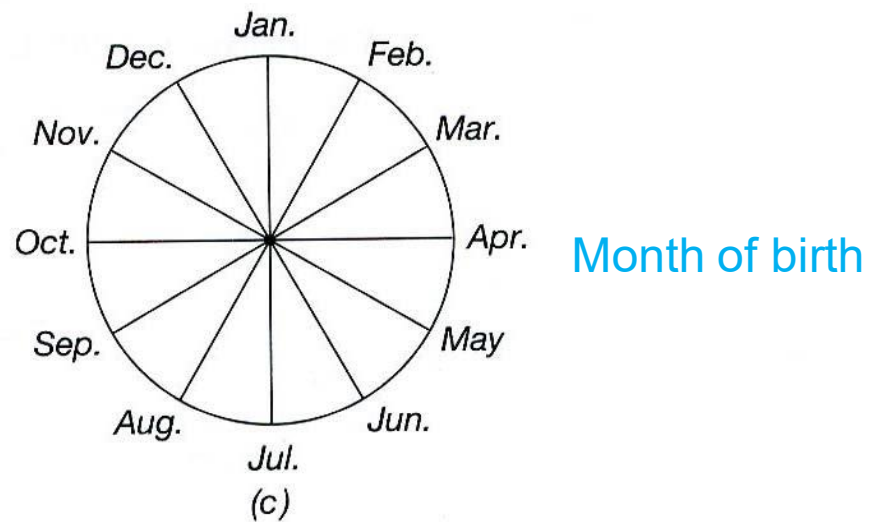
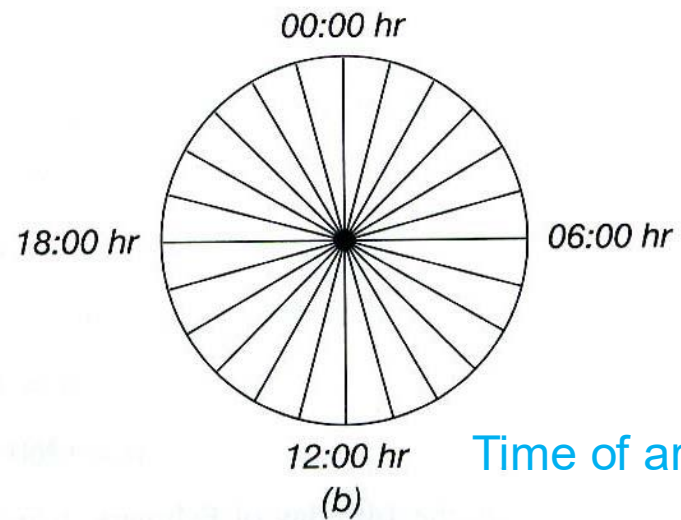
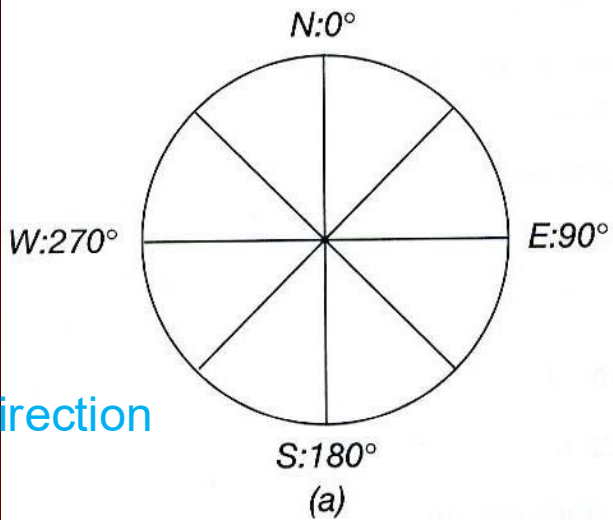
0.333

More advanced examples:

- Directional (circular) data analysis:

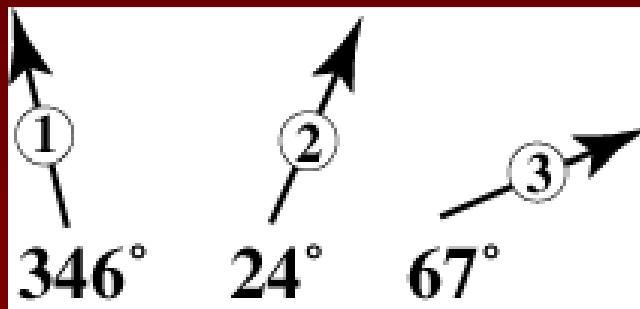
Examples of this kind of data are days of the week, months of the year, compass directions, wind directions, rotations and so on. The random variables involved are defined over the unit circle.

Some examples



Directional data cannot be treated with scalar arithmetic calculations for statistical measures because directional data are circular, and vary from 0 to 360° or 0 to 2π or 0 to 24 h.

For example, what is the average of the three directional measurements?



Treated arithmetically:

$$mean = \frac{346^\circ + 24^\circ + 67^\circ}{3} = \frac{437^\circ}{3} = 146^\circ$$



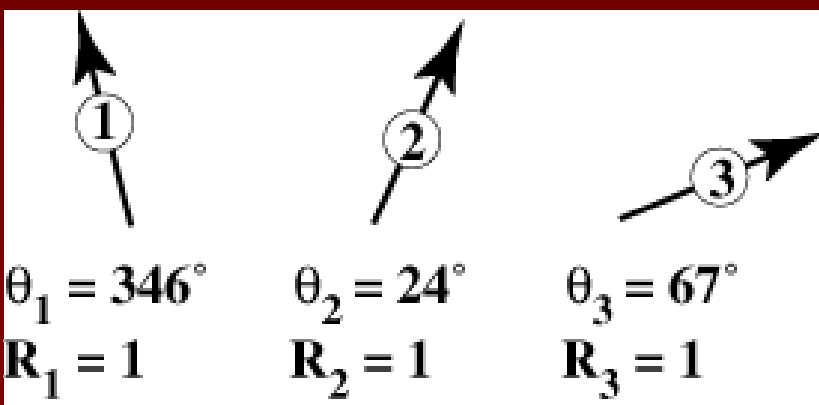
Clearly wrong!

Directional data must be treated as vectors.

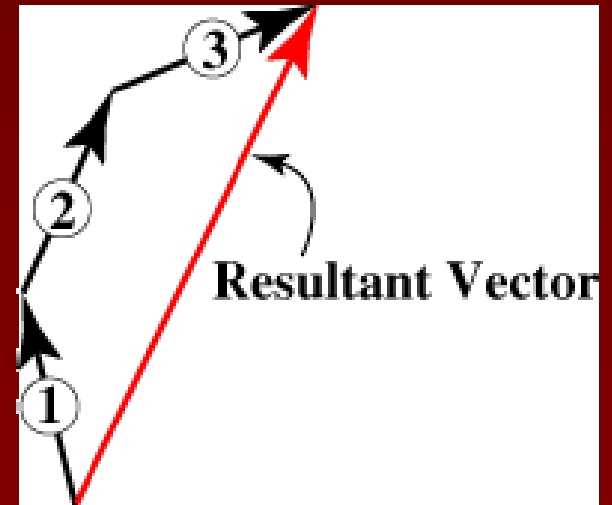
Every vector has two parts: direction and magnitude.

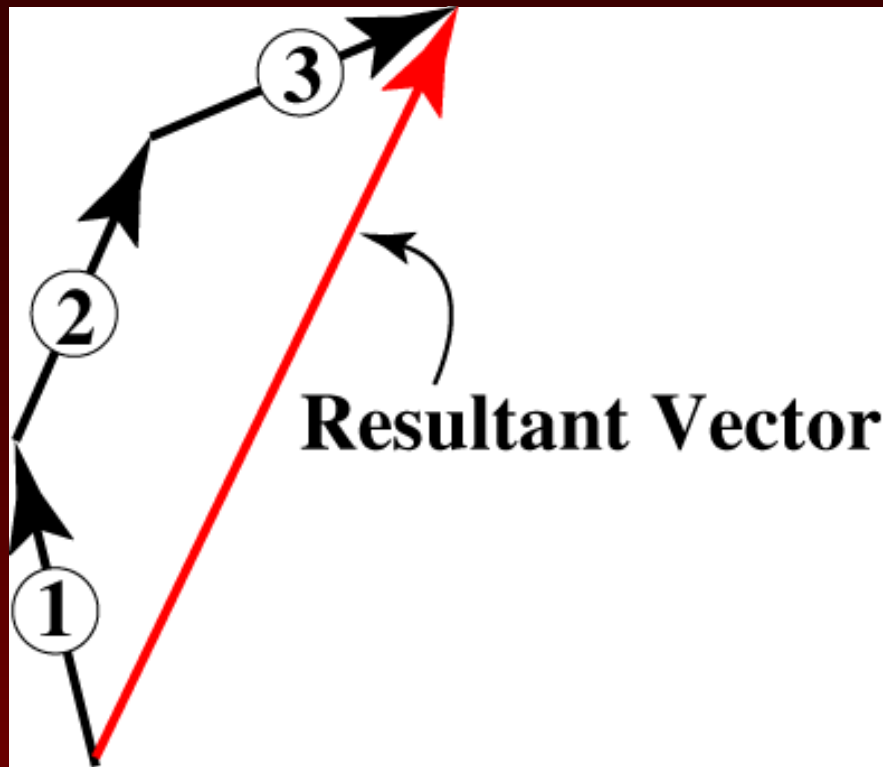
Every directional measurement is a *unit vector*; a vector with a magnitude equal to 1.

The three directional measurements are represented as:



The average direction can be determined by lining the unit vectors up end to end and joining the beginning and the end.





The “average” vector is termed the *resultant vector* .

It points in the average direction of the data.

It has a specific direction and a magnitude:

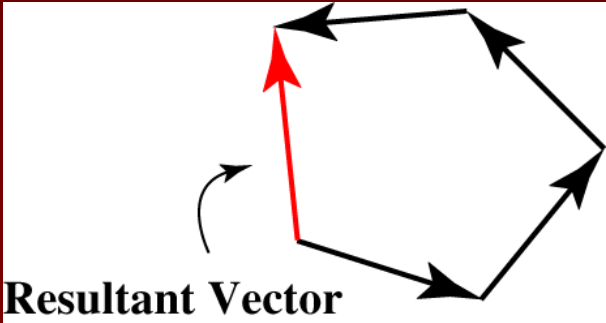
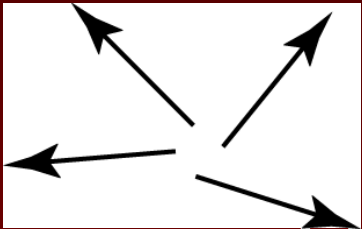
In this case:

$$\bar{\theta} = 25.5^{\circ}$$

$$R = 2.52$$

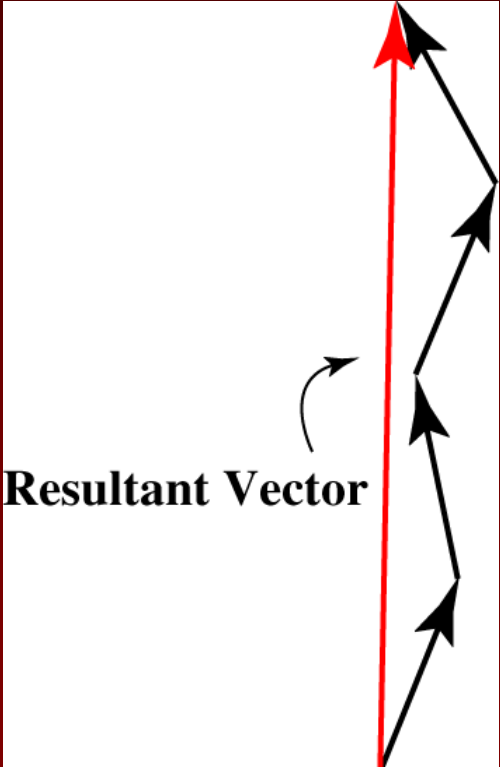
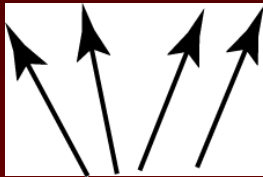
The magnitude of the resultant vector depends on the amount of variation in the directions of the directional observations.

A data set with wide dispersion of directions.



Relatively small magnitude of the resultant.

A data set with little dispersion of directions.



Relatively large magnitude of the resultant.

Circular mean and circular variance

$$A = \sum_{i=1}^n \sin(\omega \theta_i)$$

and

$$B = \sum_{i=1}^n \cos(\omega \theta_i)$$

where $\omega = \pi / 180$

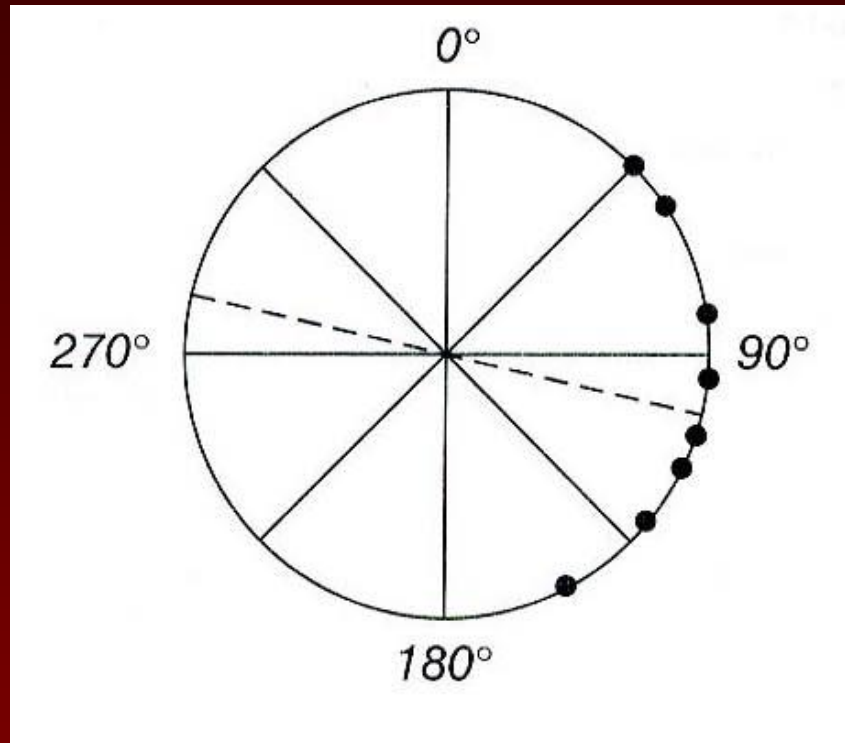
$$\bar{\theta} = \begin{cases} \omega^{-1} \arctan(A / B) & \text{if } B \geq 0 \\ \omega^{-1} \arctan(A / B) + \pi & \text{if } B < 0 \end{cases}$$

And the circular variance $\bar{R}$ is:

$$\bar{R} = 1 - \frac{1}{n} \left(A^2 + B^2 \right)^{\frac{1}{2}}$$

The scaling factor ω is 1 if the θ_i are expressed in radians

Example: directions of growth of 8 trees.



EXAMPLE 26.4 Calculating the mean angle for the data of Example 26.2.

a_i (deg)	$\sin a_i$	$\cos a_i$
45	0.70711	0.70711
55	0.81915	0.57358
81	0.98769	0.15643
96	0.99452	-0.10453
110	0.93969	-0.34202
117	0.89101	-0.45399
132	0.74315	-0.66913
154	0.43837	-0.89879
$\sum \sin a_i = 6.52069$		$\sum \cos a_i = -1.03134$
$Y = \frac{\sum \sin a_i}{n}$		$X = \frac{\sum \cos a_i}{n}$
$= 0.81509$		$= -0.12892$

$$n = 8$$

$$r = \sqrt{X^2 + Y^2} = \sqrt{(-0.12892)^2 + (0.81509)^2} = \sqrt{0.68099} = 0.82522$$

$$\cos \bar{a} = \frac{X}{r} = \frac{-0.12892}{0.82522} = -0.15623$$

$$\sin \bar{a} = \frac{Y}{r} = \frac{0.81509}{0.82522} = 0.98772$$

The angle with this sine and cosine is $\bar{a} = 99^\circ$.

The circular variance is 0.8968.

Two sample comparison.

We have two samples of the hour of arrival to the emergency unit of a hospital:

Day 1: 0.7, 21.5, 21.1, 20.5, 22.3, 23.6, 13.3, 18.8,
15.7, 16.2, 20.6, 6.4

Day 2: 13.3, 16.3, 17.5, 5.8, 23.4, 11.8, 20.4, 18.4,
11.6, 14.3, 14.6, 2.1

This is the program in R to calculate simple descriptive statistics:

```
d1<-c(0.7, 21.5, 21.1, 20.5, 22.3, 23.6, 13.3, 18.8, 15.7, 16.2, 20.6, 6.4)
d2<-c(13.3, 16.3, 17.5, 5.8, 23.4, 11.8, 20.4, 18.4, 11.6, 14.3, 14.6, 2.1)
omega<-pi/24
theta<-d1*omega
A<-sum(sin(theta));B<-sum(cos(theta))
mtime<-ifelse(B>=0, atan(A/B)/omega, (atan(A/B)+pi)/omega)
cvar<-1-sqrt(A^2+B^2)/length(theta)
mtime
[1] 17.93211
cvar
[1] 0.2961757
```

These are the statistics for d2:

```
mtime
[1] 14.43986
cvar
[1] 0.2416118
```

Procedure for a permutation test :

1- H_0 : $d1$ and $d2$ are equivalent (belong to the same population), that is, they are interchangeable.

In H_1 we can consider that the population mean times (angles) are different and also that the population circular variances are different.

2- Appropriate statistics could be the difference of sample mean times, that is the minimum between $|mt_1 - mt_2|$ and $24 - |mt_1 - mt_2|$, and the absolute value of the difference of sample variances.

3- New data sets consisting of the old data, randomly rearranged: there are $24!$ permutations. We can not be exhaustive: only an approximated test will be possible.

Code in R

```
d1<-c(0.7, 21.5, 21.1, 20.5, 22.3, 23.6, 13.3, 18.8, 15.7, 16.2, 20.6, 6.4)
d2<-c(13.3, 16.3, 17.5, 5.8, 23.4, 11.8, 20.4, 18.4, 11.6, 14.3, 14.6, 2.1)
omega<-pi/24
theta1<-d1*omega; theta2<-d2*omega
n1=length (d1); n2=length (d2); total<-n1+n2
A<-sum(sin(theta1));B<-sum(cos(theta1))
mtime1<-ifelse(B>=0, atan(A/B)/omega, (atan(A/B)+pi)/omega)
cvar1<-1-sqrt(A^2+B^2)/length(theta)
A<-sum(sin(theta2));B<-sum(cos(theta2))
mtime2<-ifelse(B>=0, atan(A/B)/omega, (atan(A/B)+pi)/omega)
cvar2<-1-sqrt(A^2+B^2)/length(theta)
difmttrue= min(abs(mtime1-mtime2),24-abs(mtime1-mtime2))
difcvartrue=abs(cvar1-cvar2)
```

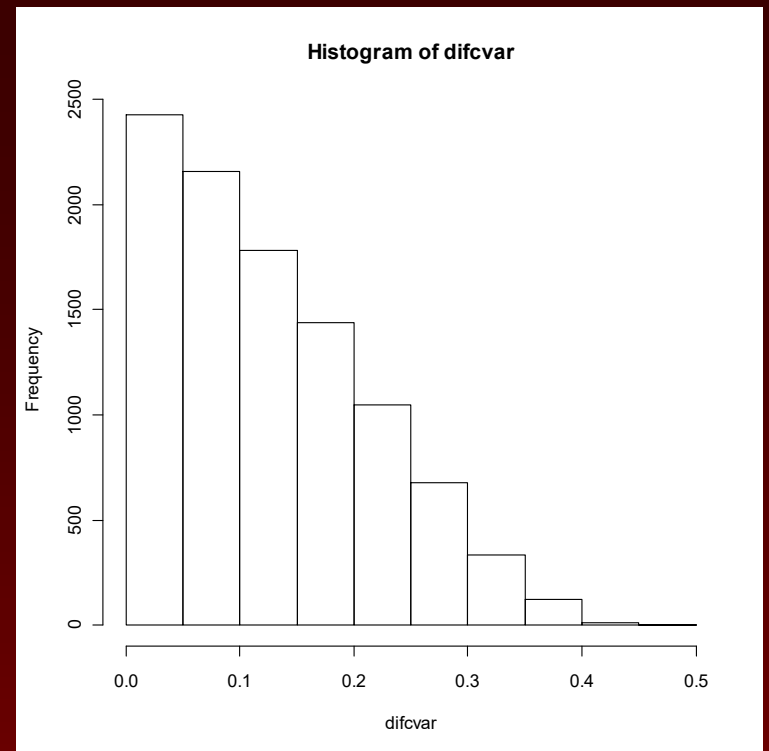
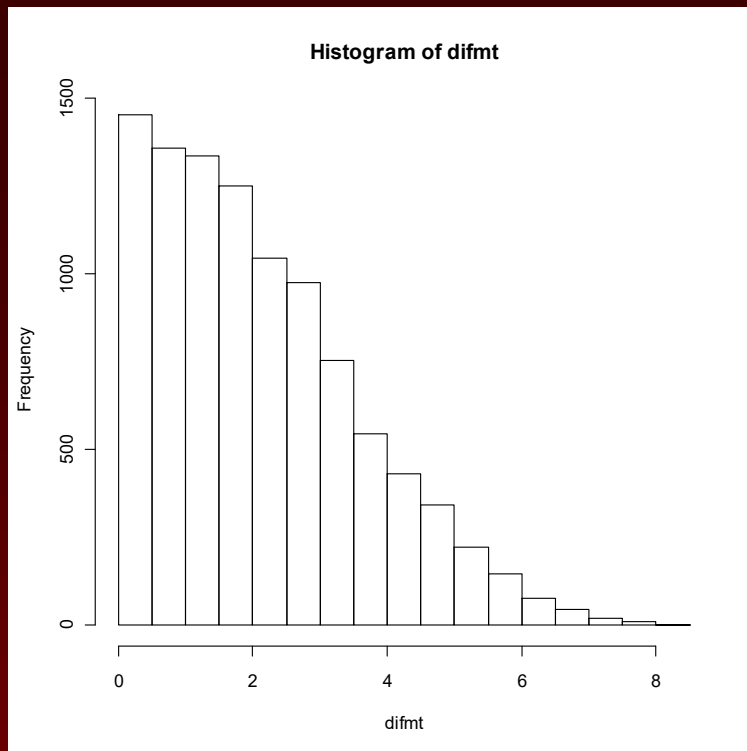
```

nr=10000 #number of rearrangements to be examined
difmt<-numeric(nr); difcvar<-numeric(nr)
vect = c(d1, d2) #Put both sets of observations in a single vector
for (i in 1:nr){
  d= sample (vect,n1+n2)
  nd1<-d[1:n1]
  a<-n1+1; nd2<-d[a:total]
  theta1<-nd1*omega; theta2<-nd2*omega
  A<-sum(sin(theta1));B<-sum(cos(theta1))
  mtime1<-ifelse(B>=0, atan(A/B)/omega, (atan(A/B)+pi)/omega)
  cvar1<-1-sqrt(A^2+B^2)/length(theta)
  A<-sum(sin(theta2));B<-sum(cos(theta2))
  mtime2<-ifelse(B>=0, atan(A/B)/omega, (atan(A/B)+pi)/omega)
  cvar2<-1-sqrt(A^2+B^2)/length(theta)
  difmt[i]= min(abs(mtime1-mtime2),24-abs(mtime1-mtime2))
  difcvar[i]=abs(cvar1-cvar2) }
length(difmt[difmt>=difmttrue])/nr
length(difcvar[difcvar>=difcvartrue])/nr

```

```
[1] 0.1844
```

```
[1] 0.7366
```



difmttrue: [1] 3.49225

difcvartrue: [1] 0.05456

There are not significant differences.